Bidual-Complemented Codomains and the Extendable Quotient Model for $\mathrm{Lip}_0(X,Y)/L(X,Y)$

Automatic math-discovery packet

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Source questions

The source paper is Anil Kumar Karn and Arindam Mandal, *Position of* $L(X,Y)$ *in* $\mathrm{Lip}_0(X,Y)$, arXiv:2506.09324. It records the following two questions.

POSITION OF $L(X,Y)$ IN $Lip_0(X,Y)$

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ABSTRACT. We prove that $L(X,Y)$ is complemented in $Lip_0(X,Y)$ (via a norm-one projection) provided that Y is a dual space. Next, we introduce a vector-valued Lipschitz-free space $F_Y(X)$, a linear contraction $\beta_X^Y : F_Y(X) \to Y$ and prove that the quotient space $Lip_0(X,Y)/L(X,Y)$ is isometrically isomorphic to $L(\ker(\beta_X^Y), Y)$ whenever Y is injective. We also consider a Y-valued duality pairing between $Lip_0(X,Y)$ and $F_Y(X)$ and obtain a necessary and sufficient condition for a Lipschitz map to be linear. As an application, we describe $\ker(\beta_{\mathbb{R}}^{\mathbb{R}})$ as the pre-dual of the quotient space $L^\infty(\mathbb{R})/\tilde{\mathbb{R}}$ where $\tilde{\mathbb{R}}$ is the set of all constant maps on $\mathbb{R}$.

1. Introduction and Preliminaries

A mapping $f : X \to Y$ between two real Banach spaces X and Y is called Lipschitz if there exists $k > 0$ in $\mathbb{R}$ such that $\|f(x_1) - f(x_2)\| \leq k\|x_1 - x_2\|$ for all $x_1, x_2 \in X$. The set of all such mappings is denoted by $Lip(X,Y)$, which forms a vector space with respect to pointwise addition and scalar multiplication. In the study of Lipschitz mappings, special attention is given to those mappings that send the 0 to 0. The set of such mappings is denoted by $Lip_0(X,Y)$. That is, $Lip_0(X,Y) = \{f \in Lip(X,Y) : f(0) = 0\}$. Further $Lip_0(X,Y)$ is a Banach space with the norm

$$Lip(f) = \sup\left\{\frac{\|f(x) - f(y)\|}{\|x - y\|} : x, y \in X, x \neq y\right\} \text{ for any } f \in Lip_0(X,Y).$$

The space $Lip_0(X,Y)$ is often regarded as a natural non-linear generalization of $L(X,Y)$, the space of bounded linear operators between X and Y. It is known that $L(X,Y)$ is a subspace of $Lip_0(X,Y)$. The space $L(X,Y)$ carries the usual operator norm,

$$\|T\| = \sup\{\|Tx\| : x \in X \text{ and } \|x\| \leq 1\}$$

for any $T \in L(X,Y)$. We have $\|T\| = Lip(T)$ for all $T \in L(X,Y)$ so that $L(X,Y)$ is a closed subspace of $Lip_0(X,Y)$. This leads to the following question:

Question 1.1. *Is $L(X,Y)$ complemented in $Lip_0(X,Y)$?*

The answer to this question is known to be affirmative in the case $Y = \mathbb{R}$. In fact, by a result due to Lindenstrauss ([5, Theorem 2]) we know that there exists a (norm one)

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Figure 1: Question 1.1 asks whether $L(X,Y)$ is complemented in $\mathrm{Lip}_0(X,Y)$.

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projection from $Lip_0(X,\mathbb{R})$ onto $L(X,\mathbb{R})$ (see also [2, Proposition 7.5]). In this paper, we extend the result and obtain an affirmative answer to Question 1.1 when Y is a dual space. We prove that there exists a surjective linear norm-one projection $P_X^Y : Lip_0(X,Y) \to L(X,Y)$. This leads us to conclude that $L(X,Y)$ is complemented in $Lip_0(X,Y)$, whenever Y is a dual space. More precisely, we prove that $Lip_0(X,Y)$ is equal to $L(X,Y) \oplus_\infty \ker(P_X^Y)$ as vector space and the two norms are equivalent.

Encouraged by first set of observations, we consider the next question:

Question 1.2. *what is the quotient space:* $Lip_0(X,Y) \big/ L(X,Y)$*?*

As $L(X,Y)$ is a closed subspace of $Lip_0(X,Y)$, $Lip_0(X,Y) \big/ L(X,Y)$ is also a Banach space. We investigate whether it is possible to characterize it again as an operator space. We prepare to answer this question in the following way.

For each $x \in X$, we define $\delta_x^Y : Lip_0(X,Y) \to Y$ by $\delta_x^Y(f) = f(x)$ for all $f \in Lip_0(X,Y)$. We show that $\delta_x^Y \in L(Lip_0(X,Y),Y)$ with $\|\delta_x^Y\| = \|x\|$ for all $x \in X$. This induces a Lipschitz mapping $\delta_X^Y : X \to L(Lip_0(X,Y),Y)$ given by $(x \mapsto \delta_x^Y)$. We consider the closed subspace

$$F_Y(X) := \overline{span\{\delta_x^Y : x \in X\}}^{\|\cdot\|}$$

of $L(Lip_0(X,Y),Y)$. (This is analogous to the Lipschitz-free space introduced by Godefroy and Kalton in [7].) We then consider a bounded linear contraction $\beta_X^Y : F_Y(X) \to X$ which we prove to be a left inverse of the Lipschitz map $\delta_X^Y : X \to F_Y(X)$. This operator leads us to provide a partial answer to the Question 1.2. We prove that the space $Lip_0(X,Y) \big/ L(X,Y)$ is isometrically isomorphic to $L(\ker(\beta_X^Y),Y)$ when Y is an injective Banach space.

Next, we examine the relation between P_X^Y and β_X^Y. We note that $\ker(P_X^Y)$ is topologically isomorphic to $Lip_0(X,Y) \big/ L(X,Y)$ which in turn can be proved to be isometrically isomorphic to $L(\ker(\beta_X^Y),Y)$, whenever Y is a dual injective space. Both the results depend on the following observation:

$$\{f \in Lip_0(X,Y) : \gamma(f) = 0 \ \ \forall \gamma \in \ker(\beta_X^Y)\} = L(X,Y).$$

We further prove that $L\left(F_Y(X) \big/ \ker(\beta_X^Y), Y\right)$ is isometrically isomorphic to $L(X,Y)$, whenever Y is a dual injective space. In fact, we prove more general versions of these results and discuss the embedding of $L(X,Y)$ into $L(F_Y(X),Y)$.

At the end, we present an explicit description of the space $\ker(\beta_{\mathbb{R}}^{\mathbb{R}})$. We prove that it is isometrically isomorphic to $\{f \in L^1(\mathbb{R}) : \int_{\mathbb{R}} f d\mu = 0\}$, where μ is the Lebesgue measure on $\mathbb{R}$. We deduce that

$$L^\infty(\mathbb{R}) \big/ \{\text{set of all constant maps on } \mathbb{R}\}$$

is isometrically isomorphic to dual of $\ker(\beta_{\mathbb{R}}^{\mathbb{R}})$.

Figure 2: Question 1.2 asks for the quotient $\mathrm{Lip}_0(X,Y)/L(X,Y)$. The source proves the usual operator-space description when Y is injective.

Source inputs

We use the following results and notation from Karn–Mandal.

1. If Z is a dual Banach space, then there is a norm-one projection
$$P_X^Z : \mathrm{Lip}_0(X, Z) \longrightarrow L(X, Z).$$
2. The vector-valued Lipschitz-free space $F_Y(X)$, the map $\delta_X^Y : X \to F_Y(X)$, and the contraction $\beta_X^Y : F_Y(X) \to X$ are defined in the source paper.
3. The map
$$\Delta_X^Y : L(F_Y(X), Y) \longrightarrow \mathrm{Lip}_0(X, Y), \qquad \Delta_X^Y(A) = A \circ \delta_X^Y,$$
is a surjective linear isometry.
4. With $D = \ker(\beta_X^Y)$, the source paper proves
$$L(X, Y) = {}^{\diamond}D := \{f \in \mathrm{Lip}_0(X, Y) : \gamma(f) = 0 \text{ for all } \gamma \in D\}.$$

Proof intuition

The dual-codomain projection of Karn–Mandal can be used one level above the desired codomain. If Y sits complemented inside a dual space Z, send a Y-valued Lipschitz map into Z, project it to a Z-valued linear operator by the invariant-mean projection, and finally project the range back into Y.

For the quotient, injectivity of Y is needed only to say that every operator $D \to Y$ extends to $F_Y(X)$ without increasing norm. Without injectivity, the same proof still identifies the quotient exactly with the operators on D that do extend, but the natural norm is the least norm of an extension.

Complemented codomains inside dual spaces

Theorem 1. *Let X and Y be Banach spaces. Suppose that $J : Y \to Z$ is an isometric embedding into a dual Banach space Z, and that $Q : Z \to JY$ is a bounded projection. Then $L(X, Y)$ is complemented in $\mathrm{Lip}_0(X, Y)$. More precisely, there is a projection*
$$\Pi : \mathrm{Lip}_0(X, Y) \longrightarrow L(X, Y)$$
with $\|\Pi\| \le \|Q\|$.

Proof. Let $P_X^Z : \mathrm{Lip}_0(X, Z) \to L(X, Z)$ be the norm-one projection supplied by Karn–Mandal for the dual codomain Z. For $f \in \mathrm{Lip}_0(X, Y)$, define
$$\Pi f = J^{-1} \circ Q \circ P_X^Z(J \circ f).$$
Here $J \circ f$ is regarded as an element of $\mathrm{Lip}_0(X, Z)$. Since $P_X^Z(J \circ f)$ is a bounded linear map $X \to Z$, and Q is a bounded linear projection onto JY, the map $\Pi f : X \to Y$ is bounded linear.

The operator Π is linear and satisfies
$$\|\Pi f\| \le \|J^{-1}\| \, \|Q\| \, \|P_X^Z\| \, \mathrm{Lip}(J \circ f) = \|Q\| \, \mathrm{Lip}(f),$$
because J is an isometry and $\|P_X^Z\| = 1$. If $T \in L(X, Y)$, then $J \circ T \in L(X, Z)$, so
$$P_X^Z(J \circ T) = J \circ T.$$
Also $Q(JTx) = JTx$ for every $x \in X$. Hence $\Pi T = T$. Thus Π is a bounded projection from $\mathrm{Lip}_0(X, Y)$ onto $L(X, Y)$. $\square$

Corollary 1. *If the canonical image of Y is λ-complemented in Y^{**}, then $L(X,Y)$ is λ-complemented in $\mathrm{Lip}_0(X,Y)$ for every Banach space X.*

Proof. Apply the theorem with $Z = Y^{**}$, $J : Y \to Y^{**}$ the canonical isometric embedding, and $Q : Y^{**} \to JY$ a projection with $\|Q\| \le \lambda$. □

The extendable quotient model

Let $F = F_Y(X)$. For a closed subspace $D \subset F$, define

$$\mathrm{Ext}_Y(D) = \{A|_D : A \in L(F,Y)\}.$$

Equip this range with the extension norm

$$\|u\|_{\mathrm{ext}} = \inf\{\|A\| : A \in L(F,Y),\ A|_D = u\}.$$

Theorem 2. *For every closed subspace $D \subset F_Y(X)$, the quotient*

$$\mathrm{Lip}_0(X,Y)/^{\diamondsuit}D$$

is linearly isometric to $(\mathrm{Ext}_Y(D), \|\cdot\|_{\mathrm{ext}})$. In particular, with $D = \ker(\beta_X^Y)$,

$$\mathrm{Lip}_0(X,Y)/L(X,Y) \cong \mathrm{Ext}_Y(\ker(\beta_X^Y))$$

linearly and isometrically, where the right side carries the extension norm.

Proof. Let $\Delta = \Delta_X^Y$. Define

$$R : L(F_Y(X),Y) \longrightarrow \mathrm{Ext}_Y(D), \qquad R(A) = A|_D.$$

By definition R is onto. Its kernel is

$$\ker R = \{A \in L(F_Y(X),Y) : A|_D = 0\}.$$

For $A \in L(F_Y(X),Y)$, the condition $A|_D = 0$ is exactly the condition that $\Delta(A) = A \circ \delta_X^Y$ lies in $^{\diamondsuit}D$. Hence

$$\Delta(\ker R) = {}^{\diamondsuit}D.$$

Since Δ is a surjective linear isometry, it induces a linear isometric identification between

$$L(F_Y(X),Y)/\ker R \quad \text{and} \quad \mathrm{Lip}_0(X,Y)/^{\diamondsuit}D.$$

The first quotient is exactly $\mathrm{Ext}_Y(D)$ with the norm

$$\|R(A)\| = \inf\{\|A+B\| : B \in \ker R\} = \inf\{\|C\| : C|_D = A|_D\} = \|R(A)\|_{\mathrm{ext}}.$$

This proves the first assertion.

For $D = \ker(\beta_X^Y)$, Karn–Mandal prove $L(X,Y) = {}^{\diamondsuit}\ker(\beta_X^Y)$. Substitution gives the stated identification of $\mathrm{Lip}_0(X,Y)/L(X,Y)$. □

Corollary 2. *If Y is injective, then*

$$\mathrm{Lip}_0(X,Y)/L(X,Y) \cong L(\ker(\beta_X^Y), Y)$$

isometrically with the usual operator norm.

Proof. Injectivity of Y says that every operator $u : D \to Y$, with $D \subset F_Y(X)$, extends to an operator $A : F_Y(X) \to Y$ with $\|A\| = \|u\|$. Therefore $\mathrm{Ext}_Y(D) = L(D,Y)$ and $\|\cdot\|_{\mathrm{ext}}$ equals the usual operator norm. Taking $D = \ker(\beta_X^Y)$ recovers the source paper's injective-codomain quotient statement. □

Limitations and novelty check

This packet does not solve the complementability problem for arbitrary codomains Y. The obstruction is exactly visible in the proof: applying the Karn–Mandal projection with codomain Y^{**} generally gives a linear operator $X \to Y^{**}$, and a projection back to Y is needed to obtain a Y-valued operator.

Cheap run-index searches found no exact previous packet for arXiv:2506.09324. Bounded web searches for the title, for $L(X, Y)$ complemented in $\mathrm{Lip}_0(X, Y)$, for dual-codomain variants, and for the quotient keywords did not reveal a later direct solution.

Human-review recommendation

Review as a partial result. The first theorem is a short functorial consequence of the source projection theorem. The second theorem is a quotient-norm reformulation of the source paper's injective-codomain argument and should be checked mainly for the exact identification of ${}^{\diamond}\ker(\beta_X^Y)$ with $L(X, Y)$.

References

[1] A. K. Karn and A. Mandal, *Position of $L(X, Y)$ in* $\mathrm{Lip}_0(X, Y)$, arXiv:2506.09324, 2025.